

DATA STRUCTURES LABORATORY

Program: Linked Queue for Student Records

Problem Statement

Question:

Write a menu-driven program to implement a linked queue for storing student records containing name, roll number, and marks. Implement the following operations:

1. **Enqueue:** Add a student record at the rear of the queue.
2. **Dequeue:** Remove a student record from the front of the queue.
3. **Display:** Show all student records in the queue.

Source Code (linked-queue.c)

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  struct student {
5      char name[50];
6      int roll_no;
7      int marks;
8      struct student *next;
9  };
10
11 typedef struct student node;
12
13 node *front = NULL;
14 node *rear = NULL;
15
16 void enqueue() {
17     node *temp = (node *)malloc(sizeof(node));
18     if (temp == NULL) {
19         printf("Memory allocation failed!\n\n");
20         return;
21     }
22
23     printf("Enter name: ");
24     scanf(" %s", temp->name);
25
26     printf("Enter roll no: ");
27     scanf("%d", &temp->roll_no);
28
29     printf("Enter marks: ");
30     scanf("%d", &temp->marks);
31
32     temp->next = NULL;
33
34     if (rear == NULL) {
35         front = rear = temp;
36     } else {
37         rear->next = temp;
```

```
38     rear = temp;
39 }
40
41 printf("[+] Enqueued successfully!\n\n");
42 }
43
44 void dequeue() {
45     if (front == NULL) {
46         printf("[-] Queue Underflow! No records to dequeue.\n\n");
47         return;
48     }
49
50     node *temp = front;
51     printf("[+] Dequeued: Roll No: %d | Name: %s | Marks: %d\n\n",
52         temp->roll_no, temp->name, temp->marks);
53
54     front = front->next;
55
56     if (front == NULL) {
57         rear = NULL;
58     }
59
60     free(temp);
61 }
62
63 void display() {
64     if (front == NULL) {
65         printf("[-] Queue is empty.\n\n");
66         return;
67     }
68
69     node *curr = front;
70     printf("\n--- Student Queue ---\n");
71     while (curr != NULL) {
72         printf("Roll No: %d | Name: %s | Marks: %d\n",
73             curr->roll_no, curr->name, curr->marks);
74         curr = curr->next;
75     }
76 }
77
78 int main() {
79     int choice;
80
81     while (1) {
82         printf("1. Enqueue\n");
83         printf("2. Dequeue\n");
84         printf("3. Display Queue\n");
85         printf("4. Exit\n");
86         printf("Choice: ");
87
88         if (scanf("%d", &choice) != 1) {
89             printf("Invalid input. Exiting...\n");
90             break;
91         }
92
93         switch (choice) {
94             case 1:
95                 enqueue();
96                 break;
97             case 2:
98                 dequeue();
99                 break;
```

```
100         case 3:
101             display();
102             break;
103         case 4: {
104             // Free any remaining nodes on exit
105             node *curr = front;
106             while (curr != NULL) {
107                 node *next_node = curr->next;
108                 free(curr);
109                 curr = next_node;
110             }
111             front = rear = NULL;
112             printf("Exiting program...\n");
113             return 0;
114         }
115         default:
116             printf("Invalid choice!\n\n");
117     }
118 }
119
120 return 0;
121 }
```

Sample Output

```
1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Choice: 1
Enter name: John
Enter roll no: 101
Enter marks: 85
[+] Enqueued successfully!

1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Choice: 1
Enter name: Alice
Enter roll no: 102
Enter marks: 92
[+] Enqueued successfully!

1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Choice: 3

--- Student Queue ---
Roll No: 101 | Name: John | Marks: 85
Roll No: 102 | Name: Alice | Marks: 92

1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Choice: 2
[+] Dequeued: Roll No: 101 | Name: John | Marks: 85
```

```
1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Choice: 3

--- Student Queue ---
Roll No: 102 | Name: Alice | Marks: 92

1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Choice: 2
[+] Dequeued: Roll No: 102 | Name: Alice | Marks: 92

1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Choice: 2
[-] Queue Underflow! No records to dequeue.

1. Enqueue
2. Dequeue
3. Display Queue
4. Exit
Choice: 4
Exiting program...
```